EXPERIMENT 1

BAR PENDULUM

AIM: Determination of the value of acceleration due to gravity (g) with the help of the Bar (Compound) Pendulum.

APPARATUS: Bar Pendulum, Knife edge, Stop Watch, Meter Scale, graph paper.

THEORY: The time-period T of the Bar pendulum is given by

$$T = 2\pi\sqrt{\frac{k^2+l^2}{lg}} \quad \text{.....} (1)$$

Where, l – the distance of the point of suspension from the centre of gravity,

k- radius of gyration of the pendulum about an axis passing through its C.G

Since the periodic time of a simple pendulum is given by $T = 2\pi\sqrt{\frac{L}{g}}$, the period of the rigid body (compound pendulum) is the same as that of a simple pendulum of length

$$L = \frac{l^2+k^2}{l} = l + \frac{k^2}{l} \quad \text{.....} (2)$$

This length (L) is known as the length of simple equivalent pendulum. The expression for L can be written as a quadratic in l. Thus from (2)

$$l^2 - lL + k^2 = 0 \quad \text{.....} (3)$$

This give two values of l ($l_1$and l_2) for which the body has equal times of vibration. From the theory of quadratic equations,

$$l_1+l_2=L$$

And,

$$l_1 l_2 = k^2$$

As the sum and the product of two roots are positive, the two roots are both positive. This means that there are two positions of the centre of suspension on the same side of the C.G., about which the periods (T) would be the same. On the other side of the C.G. similarly there will be two more point of suspension, about which the time periods (T) will again be the same. Thus, there are altogether four points, two on either of the C.G. about which the time periods of the pendulum are the same (T). The distance between two such points, asymmetrically situated on either sides of the C.G., will be the length (L) of the simple equivalent pendulum.

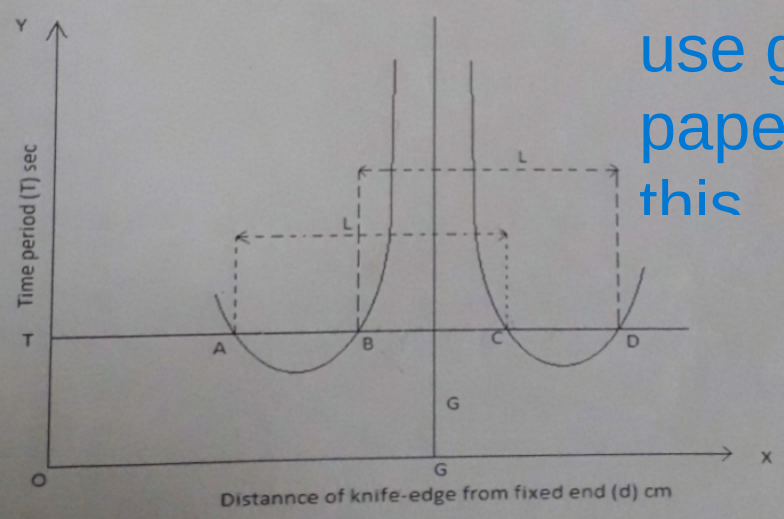
If the periods of oscillation about these points are T, then from the expression $T = 2\pi\sqrt{\frac{L}{g}}$, we get,

$$g = 4\pi^2 \frac{L}{T^2} \quad \text{.....} (4)$$

By finding **L** and **T** graphically, the acceleration due to gravity (**g**) at the place of the experiment can be found out.

PROCEDURE:

1. The bar is suspended on the knife edge by introducing it through the hole nearest to one end.
2. The bar is made to oscillate with very small amplitude and care is taken to see that the plane of oscillation is always vertical. Time taken for **20** oscillations is noted with the help of a stop watch and there from, the period of oscillation (*T*) is determined.
3. Similar operation is repeated for other successive holes on one side of the **CG** (the middle hole) and the corresponding time period for each hole is determined twice, from which the means time period (*T*) is determined.
4. The bar is now inverted and by introducing through the holes on the other side, similar operations are repeated to determine the time period corresponding to each hole.
5. The distance **(d)** of the hole from a certain fixed end **(say A)** is measured by a scale.
6. A graph is then drawn with the period (*T*) along Y-axis and the distance **(d)** of knife edge **(hole)** from a fixed end **A** along X-axis. The nature of the graph will be as shown in the figure.



use graph paper to graph this

T versus d Graph

(A) Determination of a period (*T*) and distance (d) of knife edge from one fixed end.

TABLE 1:

No. of holes from one fixed end(A)		Distance of knife edge (d) from the fixed end (cm)	Time for 20 oscillations (mins.& secs.)		Mean time taken (mins.& secs.) T'	Time period (*T*) (Secs) T = T'/20
			T_1	T_2		
On one side of C.G.	1.					
	2.					
	3.					
	4.					
	5.					
	6.					
	7.					
	8.					
	9.					
On other side of C.G.	11.					
	12.					
	13.					
	14.					
	15.					
	16.					
	17.					
	18.					
	19.					

(B) Determination of the value of *g* from graph:

TABLE 2:

For the determination of the value of *g* from the graph (1), a horizontal line is drawn parallel to

X-axis cutting the curves in four points **A, B, C, D**. The distance between the asymmetric point and either side (**AC** and **BD**) will evidently be equal to the length of the equivalent simple pendulum (***L***).The mean value of ***L*** is obtained ½ **(AC+BD)**.The periods are the same (***T***),for this four points. The value of the ordinate corresponding to the horizontal line drawn, gives the value of this common period (***T***) for the equivalent line (***L***).Putting the values of ***L*** and ***T*** in equation $T = 2\pi\sqrt{\frac{L}{g}}$, the value of *g* can be determined. Several parallel lines (horizontal) are drawn to cut the curves in four points and the mean value of *g* is obtained from the value of *g* obtained different horizontal lines.

No. of obs.	Length AC (cm)	Length BD (cm)	Length of Eq. simple pendulum $L=\frac{AC+BD}{2}$ (cm)	Corresponding Value of time period(*T*)from graph (secs.)	$g=4\pi^2\frac{L}{T^2}$ (cm/secs^2)	Mean value of *g* (cm/secs^2)
1						
2						
3						

RESULT: The value of g is found out to be .....

PRECAUTIONS:

1. The oscillations must be very small amplitude and the periods may be determined very accurately.
2. The knife edge must be kept horizontal and the oscillation must takes place in a vertical plane. Care must be taken to check rotational motion of the pendulum during oscillations.
3. It better to use telescope to know the period of oscillations, by concentrating attention on the fixed mark on the pendulum, which passes through the cross wire of the telescope when viewed through.